

1. Water Jug Problem

Problem Statement: Two jugs with capacities of 4 gallons and 3 gallons are given. Initially, both jugs are empty. The allowed operations are filling a jug, emptying a jug, and pouring water from one jug to another. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug.

2. Mars Rover Agent

Problem Statement: A Mars Rover must travel from a starting location to a target location on the surface of Mars. The rover should detect obstacles, choose a safe path, and reach the destination successfully.

3. 8-Queens Problem

Problem Statement: Place 8 queens on an 8×8 chessboard such that no two queens attack each other. No two queens should be in the same row, column, or diagonal.

4. Problem-Solving Agent

Problem Statement: Design a problem-solving agent that receives a goal as input, determines the appropriate action to achieve the goal, and displays the selected action.

5. Uniform Cost Search (UCS)

Problem Statement: A weighted graph is given with nodes and edge costs. The objective is to use the Uniform Cost Search (UCS) algorithm to find the minimum-cost path from the start node S to the goal node G.